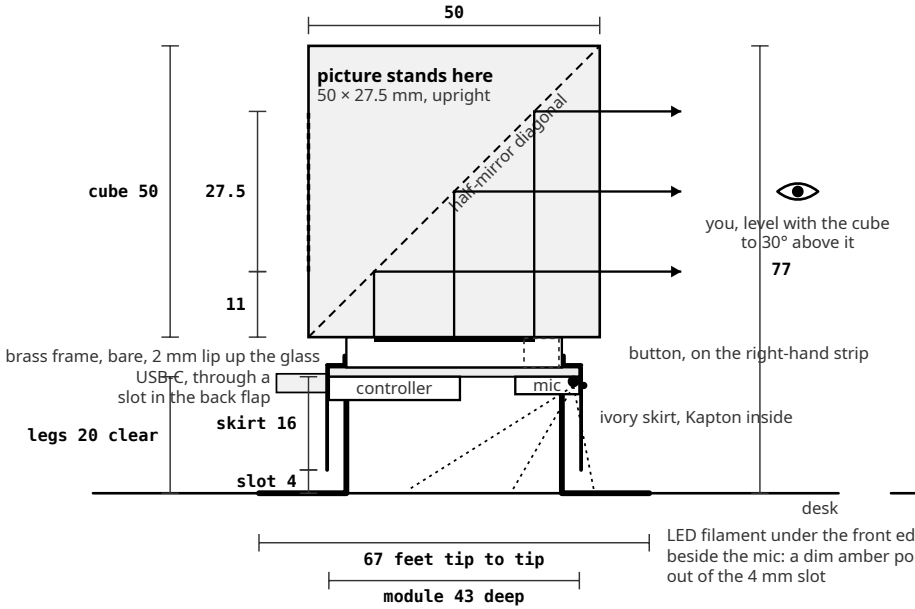
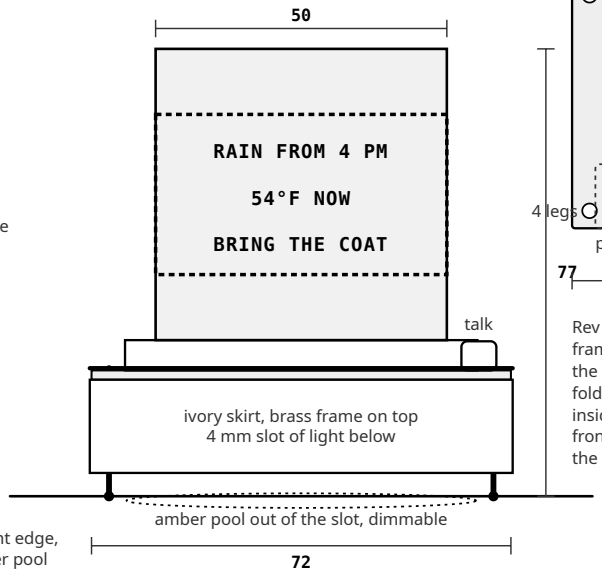
Desk hologram: how the finished thing looks

A 50 mm glass cube on the 2.42" display in a brass-topped ivory skin, four stainless legs, the answer floating inside the glass. 72 × 67 × 77 mm, about 354 g. One scale, 4 px per mm; dimensions in mm. Rev B, 2026-11-07.

Side section, viewer on the right

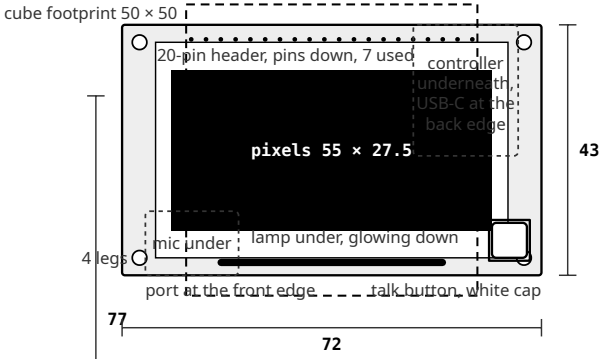


Front view: what you see



Top view, brass frame on top

module see-through; front of the desk at the bottom



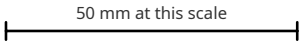
Rev B: the brass sheet lies on the board as a bare frame round the glass; its 2 mm lip turns up against the glass edge, so no green shows. The four flaps fold down 16 mm at the edges, ivory outside, Kapton inside. Holes: four legs, the button (7 × 7) at the front right, the mic port (Ø3) in the front flap, the USB-C (10 × 5) in the back flap.

Key numbers

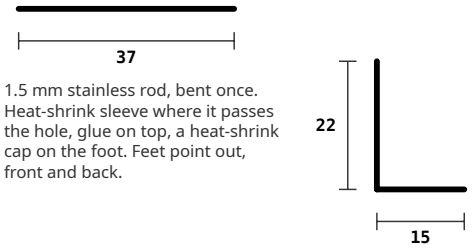
72 × 67 × 77 mm	overall, feet tip to tip
50 mm, 314 g	cube, 50:50, K9 glass
50 × 27.5 mm	picture, 116 × 64 px
3 × 14 or 8 × 19	characters a page, big or small font
4 rods, 0 solder joints	frame: legs through the module's own holes
17 wires, 2 capacitors, 1 resistor	all 30 AWG, under one PCB
3 V filament at 12 mA	amber glow under the front edge, PWM-dimmed
1 sheet of 0.13 mm brass	the skin: bare brass top, ivory skirt 16, slot 4
brass · ivory · grey · brown	the palette: frame, skirt, steel legs, the pool
5 V USB-C, under 0.4 A	power, the owned 2 A adapter
ws :8790	to the desktop's voice server over Wi-Fi

Reading the section

Every pixel sends light straight up. Half of it bounces off the diagonal and leaves through the front face, toward you.
Your eye traces it back: the picture stands at the cube's back face, the display's front edge at its top, 50 × 27.5 mm.
The other half leaks out of the top: from above you see the bare, mirrored text. From the front you see the hologram.
A black card behind the cube (optional) blacks out the room.



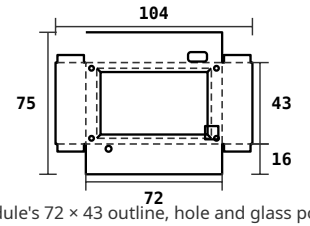
One leg, before and after the bend

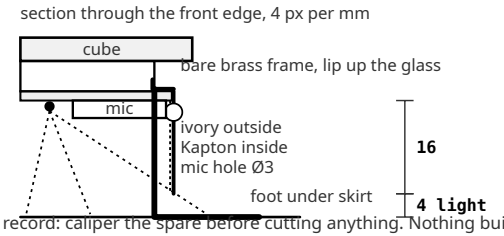


1.5 mm stainless rod, bent once. Heat-shrink sleeve where it passes the hole, glue on top, a heat-shrink cap on the foot. Feet point out, front and back.

The skin, Rev B 2026-11-07: one piece of 0.13 mm brass sheet, folded as a tray

Flat pattern 104 × 75 mm, laid along a K&S 4 × 10 in sheet. Cut the outline and the window, fold the four flaps down 16 mm, the window's 2 mm lip up, the four tabs behind the front and back flaps; solder the tabs. Scuff, wipe, mask the top at the fold, two coats of canvas white on the flaps.
Kapton on every inside face. Top: bare brass round the black glass. Edges: ivory. Below: 4 mm of light.
Card pattern first (1:1 on the build page), brass second. The skin carries nothing; the legs still do.
Right: the flat pattern (solid = cut, dashed = fold) and a section through the front edge.

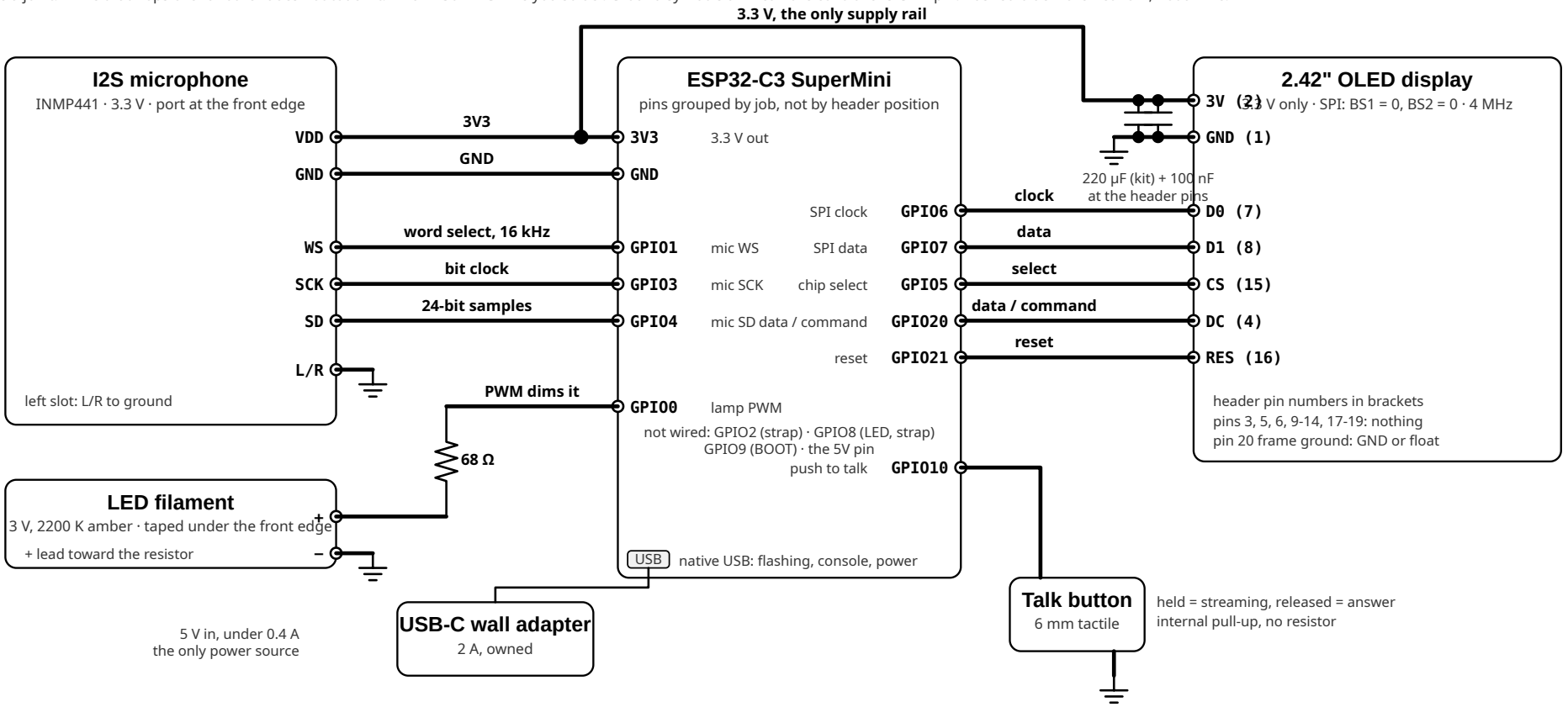




Rev B, 2026-11-07: the skin added at the builder's word (brass, ivory, grey, brown). The module's 72 × 43 outline, hole and glass positions are the Cyber Dawn record: caliper the spare before cutting anything. Nothing built.

Desk hologram: every connection on one sheet

A dot is a joint. A line that hops over another does not touch it. Thick = 30 AWG wire you solder. Ground symbols all mean the controller's GND pin. Dashed blue = the network, not a wire.



The network path: the same voice server the first Pocket AI uses, on the desk



Button down → 40 ms frames of 1280 bytes, 32 kB/s, while held. Button up → the server transcribes, asks the model, sends {"type":"transcript"} back. The cube shows it.
The server's spoken reply (PCM frames down) is discarded: this device has no speaker. Trusted home Wi-Fi only; the port is never forwarded.

Paper design 2026-11-01, not built. 0 crossing(s) on this sheet. Pin jobs: the first Pocket AI's microphone and button pins; the display on the C3's native SPI pins; no strapping pin wired.

DESK HOLOGRAM · STEP 1: THE FINAL BUILD, FOR APPROVAL

The answer, floating in a cube of glass

How it looks, how it is wired, every part and every pin. 72 × 67 × 77 mm, no soldering in the frame.
Paper design of 2026-11-01, approved the same evening (bigger cube looked into and kept at 50 mm, amber lamp added); the cube and the filament were ordered that night. **Rev B, 2026-11-07, for approval: an ivory skin with a bare brass top over the module, because its board is green and the palette for this build is brass, ivory, grey and brown.** Two things to buy for it, a sheet of brass and a tube of gel super glue. Nothing built. The physics page is Desk Hologram Physics.

What it is

A glass cube on a 2.42-inch display. Press the button, ask a question, and the answer appears as white text floating inside the cube, over a dim amber pool of light on the desk from an LED filament under the table top. The microphone's audio goes over Wi-Fi to the desktop, where the voice server that already runs for the first Pocket AI turns it into words (Whisper) and an answer (Ollama) and sends the text back.

The reference is Raspduino Uno's ESP32 OLED Hologram Lyric Video Display (August 2026, a million views): an ESP32 plays lyric frames on a 0.96-inch OLED under a beam-splitter cube. Its description names exactly three parts, and the cube is the whole trick. Ours is the same trick, four times the picture, plus ears and a brain.

This page is step 1 of the house rule: the final build first. One form sheet, one wiring sheet, every part and every pin. Rev A was approved on 2026-11-01 and its physics page followed. **Rev B (2026-11-07) changes one thing and is up for approval: the module wears a skin**, a bare brass frame on top and an ivory skirt down the edges, in this build's palette of brass, ivory, grey and brown; the wiring is untouched. Nothing is cut or soldered until it is approved.

The decisions

A CUBE, NOT A TILTED SHEET

The video's trick is a 50:50 beam-splitter cube sitting on the display. Half the light from every pixel reflects off the hidden diagonal inside the glass and leaves through the front face, so the text appears upright, inside the cube, at its back face. A 45° acrylic sheet does the same for $5 but needs a frame to hold it, a black box around it, and gives a double image from its two surfaces. The cube is one part, no frame, no ghost, 314 g of glass that stays put.

50 MM: THE LARGEST CUBE THAT SHIPS AT A SANE PRICE

The owned 2.42" display has 55 × 27.5 mm of pixels. A 50 mm cube covers 50 × 27.5 of that: 116 of 128 columns and all 64 rows. The picture hangs 11 mm above the cube's base and 39 mm below its top. At 0.43 mm a pixel, a 16-pixel line of text is 7 mm tall: three lines of 14 characters, or eight lines of 19 at the small font. Long answers page. **Bigger was asked for on 2026-11-01.** Nothing is sold between 50 and 70 mm. A 70 mm cube is $292 on Amazon (Nov 18 to 27) or $88 plus $65 shipping on eBay from China, weighs 861 g, and shows the same 55 × 27.5 mm of pixels with 10 mm more glass around them: the display, not the cube, sets the picture. A bigger hologram means a bigger display under a bigger cube (a 3.12-inch 256 × 64 OLED is 77 × 19 mm of pixels), which is a different build. **Decided the same evening: 50 mm stays.** The generator takes 70 by changing one number if that ever changes.

THE SPARE 2.42" DISPLAY, FACE UP, AS THE TABLE TOP

Cyber Dawn bought two; its wiring page already proved the pin jumpers, the 220 µF and the 4 MHz SPI limit on paper. Black pixels send nothing into the cube, which is why an OLED and not an LCD. The firmware mirrors the columns, so the bare display reads backwards and the reflection reads right; the kit's own 128 × 64 is the whole frame.

FOUR STAINLESS LEGS THROUGH THE MODULE'S OWN HOLES, NOTHING ELSE

The display module is the structure. Four 1.5 mm stainless rods go through its four mounting holes, 20 mm down to the desk, each bent into an L so the feet reach 15 mm out front and back: the 314 g cube sits on a 67 mm footprint instead of 37 and tips at 34° instead of 21°. Zero solder joints in the frame, zero rods beyond the legs. The controller and the microphone are Kapton-taped to the underside, inside the skin; the button sits on top beside the cube.

AN IVORY SKIN WITH A BRASS TOP (REV B, 2026-11-07)

The display module is a bright green board with an orange flex, the example the house colour rule names, and from above and from the side it was most of what you saw. On 2026-11-07 the builder set this build's palette at brass, ivory, grey and brown, so the module wears a skin the way the rover's boards do: one piece of 0.13 mm K&S brass sheet, lined with Kapton so no brass meets a pad, cut and folded as a tray. On top it is a bare brass frame round the black glass, with a 2 mm lip turned up against the glass so no board shows and the frame goes stiff. At the four edges it folds down into a 16 mm skirt, sprayed canvas white outside, that hides the header pins, the controller, the microphone, the filament and all seventeen wires. The skirt stops 4 mm above the desk, so the amber pool spills out of a slot of light all round. Holes: four for the legs, a 7 mm square for the button, a 3 mm hole at the microphone's port in the front flap, a 10 × 5 slot for the USB-C in the back flap. The skin carries nothing; the legs still do. About 5 g. The pattern is on this sheet and 1:1 on the build page: card first, then brass.

ESP32-C3 SUPERMINI, NOT AN ESP32-S3 DEVKIT

Thirteen are owned, it is 22.5 × 18 mm, and the first Pocket AI's voice firmware already streams this microphone to the desktop from this exact board. The three S3 DevKits belong to Cyber Dawn (one build, two spares) and the two N16R8 boards are unconfirmed. Nine pins are needed; the C3 has thirteen. GPIO2, 8 and 9 stay empty, so no strapping-pin resistor.

PUSH TO TALK, NO WAKE WORD, NO SPEAKER

A button on GPIO10 streams while held and takes the turn on release, the first Pocket AI's model: no always-on microphone, no wake-word model on a C3. The answer is text in the glass; the server's spoken reply is ignored. An amplifier is the one-block addition if a voice is ever wanted (the owned HiLetgo amp, three I2S pins).

AN AMBER GLOW UNDER THE TABLE TOP

Added 2026-11-01 at the builder's request. A 38 mm, 3 V LED filament, the kind inside an Edison bulb, Kapton-taped flat under the module's front edge beside the microphone, glowing down. GPIO0 drives it through 68 Ω: about 12 mA, a soft 2200 K pool on the desk under the glass, not a lamp. The same pin dims it, so it is also the only status light: barely there at rest, brighter while the button is held, a slow breath while the desktop thinks. One resistor, no transistor. If a real lamp is ever wanted, the owned MOSFET and the 5 V pin make it 200 mA in one more part.

THE DESKTOP SERVER THAT ALREADY EXISTS

`pocket_ai_assistant/character/server` listens on ws://<desktop>:8790/ws, takes 16 kHz PCM, runs faster-whisper `small.en` and Ollama `gpt-oss:20b` (reasoning effort low) and sends the reply back as a `transcript` message. The hologram is one more client. Start the server with `--host 0.0.0.0` so the Wi-Fi side can reach it; it stays on the home LAN, never port-forwarded.

HOW IT SITS WITH THE FAMILY

Brass, ivory, grey and brown, the builder's palette for this build (2026-11-07): the Pocket AIs' brass as the frame round the glass, the drone's bare steel as the four legs, the rover's ivory on the skirt, and the 2200 K pool as the brown. Black only in the glass, as on the rover; the green board, the orange flex and the amber tape all live under the skin. A module carrying its own electronics by its holes and pins like the Pocket AIs, one solid block of glass as the face. At 72 mm wide it stands beside the 82 mm Cyber Dawn and the 70 mm Sphere at the same scale. Nothing on it is decoration: four legs, one cube, one skin, the wires out of sight. Prototype, not product.

Pin map, ESP32-C3 SuperMini

Nine pins used of thirteen. The microphone and the button keep the first Pocket AI's pins, the display takes the C3's native SPI pins, and no strapping pin is wired, so there are no resistors at all.

PIN	JOB	GOES TO
3V3	3.3 V out of the board's regulator	display pin 2 (3V) · microphone VDD

GND	ground	display pin 1 (GND) · microphone GND and L/R · button · display pin 20 (frame) optional
GPIO6	SPI clock	display pin 7 (D0)
GPIO7	SPI data	display pin 8 (D1)
GPIO5	chip select	display pin 15 (CS)
GPIO20	data / command	display pin 4 (DC)
GPIO21	reset	display pin 16 (RESET)
GPIO1	I2S word select	microphone WS
GPIO3	I2S bit clock	microphone SCK
GPIO4	I2S data in	microphone SD
GPIO10	push to talk, active low, internal pull-up	button, other leg to GND
GPIO0	LED filament, PWM-dimmed through 68 Ω	68 Ω, then the filament's + lead; its − lead to GND
GPIO2	free: strapping pin, must float or sit high at boot	nothing
GPIO8	free: on-board LED and strapping pin	nothing
GPIO9	free: the BOOT button	nothing
5V	USB 5 V as an output	nothing

Every wire, and why

NET	FROM	TO	WIRE	WHY
3V3	controller 3V3	display 3V (pin 2), microphone VDD	30 AWG	Both loads sit on the SuperMini's own 3.3 V regulator: the display takes 50 to 75 mA lit, about 10 mA showing a few words on black, the microphone 1.4 mA. Nothing else may hang on this pin.
GND	controller GND	display GND (pin 1), microphone GND and L/R, the button's second leg, the filament's − lead	30 AWG	One ground. L/R to ground puts the microphone's 24 bits in the left I2S slot, the slot the firmware reads. Display pin 20 (frame ground) may join it or float.
LED	GPIO0	68 Ω, then the filament's + lead; − lead to GND	30 AWG	A 3 V filament on a 3.3 V pin needs the resistor: with about 2.5 V across the LEDs at this current, 68 Ω sets some 12 mA, inside the pin's 20 mA drive. GPIO0 floats for a moment at boot, so the glow may blink once; it is a lamp, not a switch, so that is harmless. The firmware pulls the pin low first thing, then dims it with PWM.
SCLK	GPIO6	display D0 (pin 7)	30 AWG	GPIO6 and GPIO7 are the C3's native SPI clock and data pins. The display is rated 4 MHz; the firmware clocks it at 4 MHz, well inside any pin's reach.
MOSI	GPIO7	display D1 (pin 8)	30 AWG	Data into the display only; it never talks back, so no MISO wire.

CS	GPIO5	display CS (pin 15)	30 AWG	Chip select, low while a frame is sent.
DC	GPIO20	display DC (pin 4)	30 AWG	Command or pixel data. GPIO20 is the C3's UART0 RX, idle high at boot, harmless on this line; the firmware's console goes over USB, not UART0.
RST	GPIO21	display RESET (pin 16)	30 AWG	GPIO21 is UART0 TX: the boot ROM chatters on it for a tenth of a second, which only resets a display that is about to be reset anyway. Keep the console on USB so it stays quiet afterwards.
WS	GPIO1	microphone WS	30 AWG	The I2S frame clock, 16 kHz, the same pin as the first Pocket AI.
SCK	GPIO3	microphone SCK	30 AWG	The bit clock moves from the first Pocket AI's GPIO2 to GPIO3 so no strapping pin is wired; that removes the 10 kΩ pull-up the first build needed. GPIO3 is a plain pin on the C3.
SD	GPIO4	microphone SD	30 AWG	24-bit samples in a 32-bit slot; the firmware enables GPIO4's internal pull-down, as on the first Pocket AI, so no resistor.
BTN	GPIO10	button, then GND	30 AWG	Push to talk, the first Pocket AI's pin. The internal pull-up holds it high; a press pulls it low, the firmware streams audio while it is held and takes the turn on release.
USB	wall adapter	controller USB-C	the cable	5 V in. Peaks: C3 radio about 300 mA under the 8.5 dBm transmit cap, display up to 75 mA, microphone 1.4 mA, filament 12 mA: under 0.4 A on a 2 A adapter.
C	display 3V (pin 2)	display GND (pin 1)	the kit's 220 µF plus a 100 nF, at the header	The module wants a stiff 3.3 V while its charge pump runs; the regulator is 40 mm of wire away.

Parts

Two things were bought on 2026-11-01 (the cube and the filament, logged in `docs/INVENTORY.md`). Everything else is in a drawer already, so it is listed here only so it is not bought twice. Each part is named by what it is, with the link on the word to the exact product.

TO BUY FOR REV B (2026-11-07): A SHEET OF BRASS AND A TUBE OF GLUE

QTY	PART	WHERE	PRICE	WHY AND WHAT TO CHE

1 of 6	brass sheet, 0.005 in (0.13 mm), 4 × 10 in	K&S 250, Amazon B0006N6WLM, a 6-pack; single sheets at True Value	about $3 a sheet; the rover's list carries the same line	**To bu** The s One patte 104 × mm o the 1 254 r shee laid a its le with room besic for a seco try. C with sciss or th craft knife folds a ste rule, takes solde the f corn tabs the owne 63/3 flux (baki soda alcoh after, the p bliste Scuff flaps 400 g and w with alcoh befor paint spray offcu first a chec paint does lift u a finge the g the Pock

				AI's f note beca no pr is ow
1 tube	gel super glue, 4 g	Loctite Ultra Gel Control: the one in the Amazon Saved for later, or Zoro	about $6; the rover's list carries the same tube	**To bu** The f legs i their holes drop top o each, over heat-shrin sleev and t butto the modu top. I glue the h glue is ow (benc recor 2026 07), a hot g lets g cold meta Rev A "own benc stock was wron abou Brass brass skin's corne tabs, solde not glued

BOUGHT 2026-11-01: THE ONLY TWO PARTS NOT ALREADY OWNED

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

1	beam-splitter cube, 50 mm, 50:50, K9 glass	Amazon B0B34D4H18, size 50mm (child listing B0B34FV475)	$74.99, free delivery, due 2026-11-08	**Bought 2026-11-01**, logged in INVENTORY.md. The hologram. The listing the video links to, in its largest size; it is the same cube in 10, 20, 25, 30, 40 and 50 mm. 50 mm covers 50 of the display's 55 mm of pixels and all 27.5 mm of their height, so the picture inside the glass is 50 × 27.5 mm, 116 × 64 pixels. A 40 mm cube ($48.60 from a sister listing) shows 93 columns; a 30 mm one ($29.99) only 70, too narrow for a sentence. 314 g of glass: it will not slide. Surface quality 60/40 is fine for a display; a lab-grade cube costs $400 and looks the same from a chair. **Bigger, asked for 2026-11-01:** nothing is sold between 50 and 70 mm. A 70 mm cube is $88 plus $65 shipping from China on eBay, or $292 on Amazon (Nov 18 to 27); 861 g; it shows the same 55 × 27.5 mm of pixels inside 10 mm more glass each side.
1 of 10	LED filament, 3 V, 38 mm, 2200 K amber	Amazon B0FR35BGYL (QIROGKC 10-pack; size 38MM, colour 2200k)	$9.99, due 2026-11-02	**Bought 2026-11-01**, logged in INVENTORY.md. The builder's example listing of 2026-11-01. A ceramic strip of LEDs, the kind inside an Edison bulb, with both leads at one end and a polarity mark. Rated 3 V and 200 lm flat out; here it runs at about 12 mA from GPIO0 through 68 Ω, a dim 2200 K pool on the desk, not a lamp. Ten in the pack, one used. Kapton-tape it flat to the module's underside along the front edge beside the microphone, glowing side down, leads toward the controller. The 54 and 68 mm sizes also fit under the 72 mm module if a longer line of light is wanted.

OPTIONAL, CHEAP, NOT OWNED

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK

1	black matte card, about 60 × 60 mm	any art shop (Blick, Bond St) or a black mat-board offcut	under $3	**Optional.** Stood behind the cube it turns the room behind the glass black, so the text hangs on nothing. Try the cube without it first; a dark wall does the same job.
1	5-minute epoxy	hardware store	about $6	**Optional.** For the four legs in the display module's mounting holes, if the gel super glue plus a heat-shrink sleeve does not feel solid. Epoxy fills the gap between a 1.5 mm rod and a 2.5 to 3 mm hole.

OWNED: ELECTRONICS

QTY	PART	WHERE	PRICE
1 of 2	2.42" OLED display kit, 128 × 64, SPI	Adafruit 2719	Cyber Dawn's spare, bought 2026-10-19 (cyber_clock_v2
1 of 13	ESP32-C3 SuperMini controller	Amazon B0F888JQ91 (Meshnology 10-pack) and a 3-pack	Pocket AI stock, 2026-09-02 and 2026-10-17

1 of 5	I2S microphone	Amazon B092HWW4RS (AITRIP INMP441 5-pack)	Pocket AI stock, 2026-10-23
1	tactile button, 6 × 6 mm	QTEATAK kit, owned ×420 · no link recorded	Pocket AI stock
1 + 1	220 µF electrolytic (in the display kit) and a 100 nF ceramic	BOJACK ceramic kit, Amazon B07P7HRGT9	Pocket AI stock
1	68 Ω resistor	LuminologyPro kit, owned · no link recorded	Pocket AI stock
about 0.5 m	30 AWG silicone wire	CBAZY kit, Amazon B073RDGTPB	Pocket AI stock

	Kapton tape, heat-shrink, header pins, solder, flux, wick	ELEGOO Kapton B072Z92QZ2; Pointool heat-shrink and the rest, no links recorded	owned bench stock
1	USB-C wall adapter, 2 A, and a data cable	owned, no link recorded	owned

OWNED: THE FINISH

QTY	PART	WHERE	PRICE	WHY, AND WHAT TO CHECK
	gloss canvas white spray paint	Rust-Oleum 7789830, Pip's Amazon #12, 2026-09-21	owned	**Owned.** The ivory, as on the rover: two light coats on the outside of the skirt, twenty minutes apart, with the top frame masked at the fold with Kapton. Outdoors or at an open window, never near the iron; dry overnight before the Kapton lining goes in. The satin white 7791830 is owned too; the rover uses the gloss, so the gloss.
	wet/dry sandpaper, 400 to 3000 grit	BAISDY 45-sheet pack, Pip's stock	owned	**Owned.** 400 keys the flaps for paint and deburrs every cut edge; 800, 1500 and 3000 wet bring the bare top frame up bright. No lacquer is owned: bare brass darkens slowly, like the rest of the family's brass, and alcohol and 3000 grit bring it back.
1 sheet	white cardstock, 65 lb	Amazon Basics, Pip's stock	owned	**Owned.** The pattern: print the 1:1 page, glue it to the card, cut, fold, and try the card tray on the module before any brass is cut. The card is the template only, never the finish.
1	white button cap	from the QTEATAK button kit	owned	**Owned.** Reads ivory beside the brass; the bare button is black.

OWNED: FRAME

QTY	PART	WHERE	PRICE

1 of 10	1.5 mm stainless rod, 150 mm	uxcell 304, Amazon B082ZNMD46	bought 2026-10-27 as coupon stock for the projects after Cyb

What the firmware port needs

Not written yet; listed so the pin map above is known to be enough.

- Start from `pocket_ai_assistant/character/firmware` (ESP-IDF 6.0.2, the C3 build): keep Wi-Fi, the WebSocket client, the 16 kHz I2S capture and the button logic; drop the amplifier (already a Kconfig switch).
- Replace the I2C SSD1306 panel with the same esp_lcd SSD1306 driver on SPI (SSD1309 takes the same commands): pins 6/7/5/20/21, 4 MHz, 128 × 64, segment remap flipped so the reflection reads right.
- Move the bit clock define from GPIO2 to GPIO3; console on USB Serial/JTAG, not UART0.

- Render `transcript` messages as pages of text inside the middle 116 columns; show `listening` and `thinking` as a one-word state; discard the server's audio frames.
- LEDC PWM on GPIO0 for the filament: pin low at boot, about 5 % at rest, 30 % while listening, a slow breath while thinking. The glow is the only status light.
- Ask the server's character for two-sentence answers; the glass holds about 40 words a page.

Before power, with the meter

- [ ] Display header, before the controller is powered: pins 1 and 2 (GND and 3V) not shorted. A chirp that stops is the 220 µF charging.
- [ ] Each of the five display signal wires beeps to its controller pin and to nothing next to it.
- [ ] The legs' heat-shrink sleeves isolate the stainless from the module's pads: rod to GND and rod to 3V open.
- [ ] The skin is not a conductor, before power and with it fitted: brass to GND open, brass to 3V open, brass to each leg open. The Kapton lining is what keeps them so; a beep means a lifted edge. Everything else is a supplier spec and is not metered.

After approval

Step 2, the physics page, exists for Rev A and still holds: the skin changes no optics, no wiring and 5 g of mass. After Rev B's yes comes the build day page: the bench list, the display's jumper rework, the header, the seventeen wires pin to pin, the flash, the skin, the legs, the cube, with the pass for each step.

Built by `docs/pages/gen/build_page.py`, which also writes `docs/FINAL_BUILD.md` and the PDFs; the pin map, nets, parts and dimensions live in that script. The drawing-clearance check passes. Sources read 2026-11-01: the video's description and its three listings, the Adafruit product page and learn guide for the display, the WiseChip panel drawings, the two owned inventories, `pocket_ai_assistant/character/PROTOCOL.md`. Nothing measured.